Carbon and its Compounds — MCQ Worksheet

Science • Chapter 4 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. Carbon forms a large number of compounds mainly because of:

(A) Tetravalency and catenation
(B) High atomic number
(C) Metallic nature
(D) Inert nature

Q2. The valency of carbon is:

(A) 1
(B) 2
(C) 3
(D) 4

Q3. The bonds in carbon compounds are mostly:

(A) Ionic
(B) Covalent
(C) Metallic
(D) Hydrogen

Q4. Saturated hydrocarbons are also called:

(A) Alkenes
(B) Alkynes
(C) Alkanes
(D) Aromatic compounds

Q5. The general formula of alkanes is:

(A) C■H$_2$■
(B) C■H$_2$■■$_2$
(C) C■H$_2$■■$_2$
(D) C■H■

Q6. Ethene has the formula:

(A) C_2H_2
(B) C_2H_4
(C) C_2H_6
(D) CH_4

Q7. The functional group –COOH is called:

(A) Aldehyde
(B) Ketone
(C) Carboxylic acid
(D) Alcohol

Q8. The IUPAC name of CH_3CH_2OH is:

(A) Methanol
(B) Ethanol
(C) Propanol
(D) Methanal

Q9. A homologous series differs by:

(A) $–CH_3$
(B) $–CH_2–$
(C) –OH
(D) –COOH

Q10. Burning of saturated hydrocarbons produces:

(A) Sooty yellow flame
(B) Clean blue flame
(C) No flame at all
(D) Green flame

Q11. Unsaturated hydrocarbons undergo:

(A) Substitution
(B) Addition
(C) Elimination only
(D) No reactions

Q12. The catalyst used in hydrogenation of vegetable oils is:

(A) Pt
(B) Ni
(C) Fe
(D) Cu

Q13. Methane reacts with chlorine in sunlight to give methyl chloride. This is a:

(A) Combination reaction
(B) Substitution reaction
(C) Addition reaction
(D) Decomposition reaction

Q14. Glacial acetic acid is:

(A) Dilute ethanoic acid
(B) Pure CH_3COOH
(C) Aqueous solution of HCl
(D) Mixture of acid and water

Q15. Vinegar is approximately:

(A) 100% acetic acid
(B) 5–8% acetic acid
(C) Pure ethanol
(D) Pure water

Q16. Esterification reaction is:

(A) Alcohol + Acid → Ester + Water
(B) Alcohol + Water → Ester
(C) Ester + Water → Alcohol + Acid
(D) Acid + Acid → Ester

Q17. The product formed when ethanol reacts with sodium is:

(A) Sodium ethoxide + H_2
(B) Sodium acetate
(C) Ethyl acetate
(D) Methane

Q18. Ethanoic acid + sodium bicarbonate gives:

(A) Salt + water
(B) Salt + water + CO_2
(C) Only CO_2
(D) Ester

Q19. The hydrophobic part of a soap molecule is:

(A) The $–COO^-$ end
(B) The long hydrocarbon chain
(C) The Na^+ ion
(D) The water layer

Q20. Soaps form micelles in water. Inside a micelle:

(A) Heads point inward
(B) Tails point inward and trap oily dirt
(C) All ions are arranged randomly
(D) There is no oil at all

Q21. Hard water contains ions of:

(A) Na^+ and K^+
(B) Ca^{2+} and Mg^{2+}
(C) Fe^{3+} only
(D) H^+ only

Q22. Synthetic detergents work in hard water because:

(A) They form scum easily
(B) Their Ca/Mg salts are soluble
(C) They are stronger acids
(D) They contain soap

Q23. The IUPAC name of CH_3COOH is:

(A) Methanoic acid
(B) Ethanoic acid
(C) Propanoic acid
(D) Butanoic acid

Q24. Diamond and graphite are:

(A) Different compounds of carbon
(B) Allotropes of carbon
(C) Isotopes of carbon
(D) Mixtures of carbon

Q25. The number of covalent bonds in a methane molecule is:

(A) 2
(B) 3
(C) 4
(D) 5

— End of Worksheet —